AUDIENCE SENSITIVITY, CONTENT COMPLIANCE & FORMAT PREFERENCE ANALYSIS

An Exploratory Analysis of France Top 50 Playlist Records

Research Paper

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Analytical platform	Microsoft Power BI
Dataset period	18 May 2024 – 27 November 2025
Final analytical dataset	27,699 playlist-record observations across 554 dates
Project context	France Top 50 playlist analytics case study

Abstract. This study examines a cleaned collection of France Top 50 playlist observations to describe the representation of explicit and non-explicit-flagged tracks, release-format mix, rank-related content composition, track duration, and Spotify popularity metadata. The final analytical table contains 27,699 records across 554 dates from 18 May 2024 to 27 November 2025. Explicit-flagged records account for 56.27% of all records, while album releases account for 52.88%, singles 47.09%, and compilations 0.03%. Within the Top 10 observations, 62.47% are explicit-flagged and 37.53% are clean/not flagged explicit. Average track duration is 3.09 minutes. The analysis uses descriptive statistics and Power BI visuals; it does not infer causation, individual listener attitudes, actual streams, or the commercial impact of explicit content. The findings are intended to support structured editorial review, playlist pitching, and content-compliance workflows rather than to prescribe a universal release strategy.

Keywords: music analytics; playlist analysis; explicit content; release format; Spotify popularity; Power BI; descriptive analytics; France market

1. Introduction and Research Context

Digital music catalogues contain both creative content and metadata that can support operational decisions. For playlist teams, labels, distributors, and content operations, the same track may need to be evaluated through several lenses: its presence in a chart, its recorded popularity score, its release format, its duration, and its explicit-content flag. Looking at these attributes together can help teams establish a repeatable review process. However, these fields describe different things and should not be treated as interchangeable measures of audience demand.

The present project focuses on records associated with a France Top 50 playlist over a period of approximately eighteen months. A multi-date dataset enables a view of recurring playlist positions rather than a single chart snapshot. Because a track can appear on many dates, each row is best understood as a *playlist-record observation*, not necessarily a unique song. This distinction matters when interpreting counts: a high record count can reflect repeated appearances over time, not a large number of unique tracks.

1.1 Problem statement

Music teams may need to balance editorial fit, content classification, release metadata, and track-level performance indicators. A descriptive report can make the available evidence easier to review, but only if it clearly defines the unit of analysis and avoids claiming more than the dataset supports. This project therefore evaluates how content flags and release formats are represented in playlist records, while identifying where the available dashboard does not support a firm numerical conclusion.

1.2 Aim and objectives

The main aim is to build a transparent, interactive analytical view of France Top 50 playlist records. The objectives are to: (1) validate the structure and quality of the dataset; (2) quantify explicit-flag representation overall and within the Top 10; (3) compare record counts across album, single, and compilation formats; (4) explore duration and popularity measures without assuming causation; (5) organize results into a multi-page Power BI report; and (6) translate findings into cautious operational recommendations.

1.3 Research questions

RQ	Question
RQ1	What proportion of playlist-record observations are flagged explicit versus clean/not flagged explicit?
RQ2	How does the explicit-flag composition look within the Top 10 observations?
RQ3	How are playlist records distributed across album, single, and compilation release types?
RQ4	What can the available duration and popularity visuals reasonably tell us?
RQ5	What data-quality and measurement limitations should guide business interpretation?

2. Dataset Scope and Analytical Unit

The dataset used for this report is the cleaned France Top 50 playlist table loaded into Power BI. The date field spans 18 May 2024 through 27 November 2025. After the cleaning decisions described in Section 3, the analysis contains 27,699 rows and 554 distinct dates. The dataset has approximately 50 observations per date, with 49 records on one date and 50 on the remaining dates.

2.1 Core fields

Field	Analytical role	Interpretation
date	Time dimension	Date associated with a playlist snapshot/observation.
position	Rank	Position within the Top 50; lower numeric values represent higher rank.
song	Track label	Track name used for display and identification; titles alone may not uniquely identify recordings.
popularity	Track-level score	A 0–100 Spotify popularity metadata field; it is not a direct stream count.
is_explicit	Content flag	Boolean explicit-content metadata. A false flag should be described as clean/not flagged explicit, not as an independent lyrical audit.
album_type	Release format	Categorical label: album, single, or compilation.
total_tracks	Release structure	Number of tracks associated with the release metadata.
duration_minutes	Track length	Duration converted to minutes from the source duration field.

2.2 Unit of analysis and repeated observations

The principal unit is the row-level playlist observation. Since the same song can appear on multiple dates, the total of 27,699 should not be described as 27,699 unique songs. Similarly, the 56.27% explicit share is a share of cleaned playlist records, not necessarily a share of unique tracks in the catalogue. A future version should include a stable track identifier and report both record-weighted and unique-track-weighted statistics.

2.3 Why the popularity field needs careful wording

Spotify documentation describes track popularity as a 0–100 score that is influenced substantially by play counts and recency, and notes that it may lag actual popularity. The field is therefore a platform-provided indicator, not a direct measurement of streams, audience sentiment, or commercial revenue [1]. The report uses the term *Spotify popularity score* wherever possible to preserve that distinction.

3. Data Cleaning and Methodology

The cleaning process aimed to make dates, ranks, durations, content flags, and release-format categories suitable for descriptive analysis. Cleaning decisions are documented here because they affect the final population and must be visible to anyone attempting to reproduce the dashboard.

3.1 Cleaning actions and validation

Check / action	Result or decision
Initial table size	27,800 rows × 10 columns.
Date parsing	Converted the date field to a datetime type using day-first parsing; final range: 2024-05-18 to 2025-11-27.
Exact duplicate rows	19 exact duplicates were removed.
Date with duplicate rank positions	2025-03-01 contained 81 rows but only 50 unique rank positions, with 31 duplicated positions. All rows for that date were excluded.
Missing song / zero duration record	A record dated 2025-02-02 at position 2 had a missing song and zero duration; it was removed/cleaned from the analytical table.
Final rows and dates	27,699 rows across 554 dates.
Missing values	No remaining nulls were reported in the cleaned table.
Position validation	No positions outside the expected 1–50 range were reported.
Duration validation	No zero or negative durations remained.
Popularity validation	No values outside the 0–100 range remained.

3.2 Analytical calculations

Explicit-record share = explicit-flagged records ÷ all cleaned records × 100. **Clean/not-flagged share** = records not flagged explicit ÷ all cleaned records × 100. **Top 10 clean share** = clean/not-flagged observations with position ≤ 10 ÷ all observations with position ≤ 10 × 100. **Single-to-album record ratio** = single records ÷ album records. These measures are record-weighted and therefore incorporate repeated appearances over dates.

3.3 Power BI implementation

The final dashboard uses multiple report pages to separate executive overview, content/compliance analysis, release-format structure, duration/popularity analysis, and rank-tier analysis. Power BI supports interactive reports with visualizations, filters, slicers, and cross-filtering [2]. For reliable interpretation, all measures should use consistent filters and explicitly state whether they are based on row counts, distinct tracks, averages, or percentages.

Important reproducibility note: the report excludes every record dated 1 March 2025 because the date contained conflicting rank-position entries. The resulting time series has 554 dates rather than a complete, perfectly uniform daily history. This is a material data limitation, not a cosmetic cleaning choice.

4. Overall Dataset Profile and Content Classification

The strongest overall finding is that explicit-flagged observations make up a majority of the cleaned playlist records. Of 27,699 records, 56.27% are flagged explicit and 43.73% are clean/not flagged explicit. The clean-to-explicit record ratio is reported as 0.78, meaning that there are roughly 0.78 clean/not-flagged observations for every explicit-flagged observation.

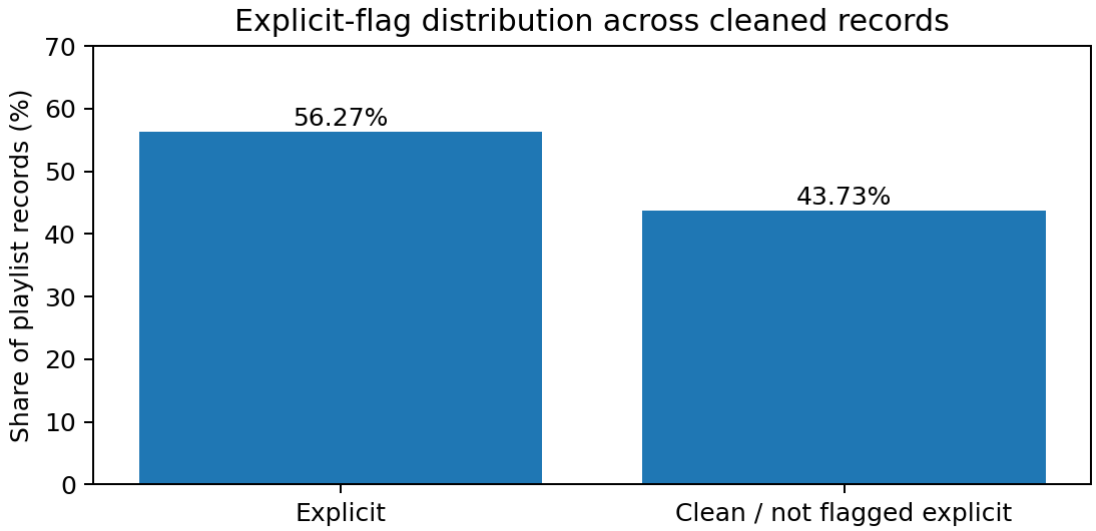


Figure 1. Overall explicit-flag distribution, based on the dashboard KPI percentages.

4.1 What this pattern means

The result establishes that explicit-flagged content is common in the observed playlist records during the study period. It does not establish that explicit content is preferred by listeners, that explicit lyrics cause higher rankings, or that clean tracks are commercially weaker. Those conclusions would require additional evidence, such as unique-track analysis, time-aligned streams, audience demographics, campaign exposure, and a suitable comparison design.

4.2 Compliance interpretation

For content operations, the share is large enough to justify a consistent explicit-content review step in playlist preparation and reporting. A useful workflow would identify the flag, verify the correct track version, confirm the intended audience and market requirements, and retain a record of any editorial decision. This is a process recommendation rather than a claim that the current content mix is non-compliant.

4.3 Metadata caveat

Spotify's track documentation describes the explicit field as indicating explicit lyrics, but also notes that a false value can mean either non-explicit or unknown [1]. Accordingly, this paper uses the phrase *clean/not flagged explicit* where precision matters. The dataset flag should be treated as a screening signal, not a substitute for manual content review where legal, contractual, or brand-safety requirements apply.

5. Rank Position and Top 10 Content Composition

The Top 10 analysis reports that 62.47% of Top 10 playlist-record observations are flagged explicit and 37.53% are clean/not flagged explicit. The dashboard's Top 10 visual represents approximately 5,539 observations across the available dates; the percentages are the more reliable figures to communicate because the dashboard displays rounded record counts.

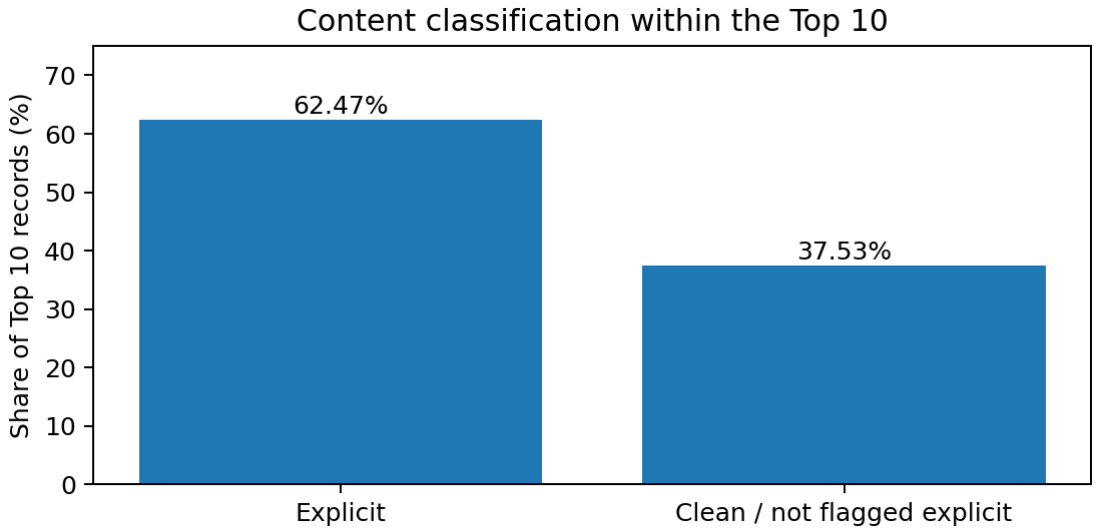


Figure 2. Content-flag composition among observations ranked in positions 1–10.

5.1 Interpretation

The Top 10 share is directionally consistent with the overall dataset: explicit-flagged observations form the larger share in both views. This is a descriptive association between rank-band membership and the content flag. It should not be presented as proof that explicit content improves ranking. The dataset contains repeated observations of tracks across dates, and rank is influenced by many factors not measured here, including artist reach, release timing, promotion, playlist/editorial decisions, and current listening activity.

5.2 Rank bands and chart-reading caution

The dashboard includes content composition by rank band and a Top 10/Top 25/Top 50 comparison. In the exported report, the rank-tier chart does not provide sufficiently legible numeric labels to support a precise percentage-by-tier table. For that reason, this paper does not invent or estimate exact values for the Top 25 or Top 50 bars. The defensible numeric statements are the overall explicit share (56.27%) and the Top 10 composition (62.47% explicit-flagged; 37.53% clean/not flagged explicit).

5.3 Operational implication

Teams can use rank-band content summaries to plan editorial checks—for example, reviewing explicit-flagged entries in the Top 10 before a playlist is promoted in a family-oriented or brand-sensitive context. The same procedure can be applied to any rank range. A policy decision should be based on the actual audience, platform controls, campaign brief, and market requirements, not on rank alone.

5.4 Recommended next analysis

Calculate explicit share by rank band from the underlying cleaned table and publish exact percentages with denominators. Add a unique-track view alongside the record-weighted view. A track that remains in the chart for many dates can otherwise contribute many times to the result.

6. Release Format and Album Structure

Release-format counts show that album-associated records form the largest category, followed closely by single-associated records. The cleaned dataset contains 14,647 album records (52.88%), 13,043 single records (47.09%), and 9 compilation records (0.03%). The single-to-album record ratio is 0.89. These values describe how the release metadata is represented in playlist observations, not necessarily the number of distinct releases or the proportion of songs originally launched as albums versus singles.

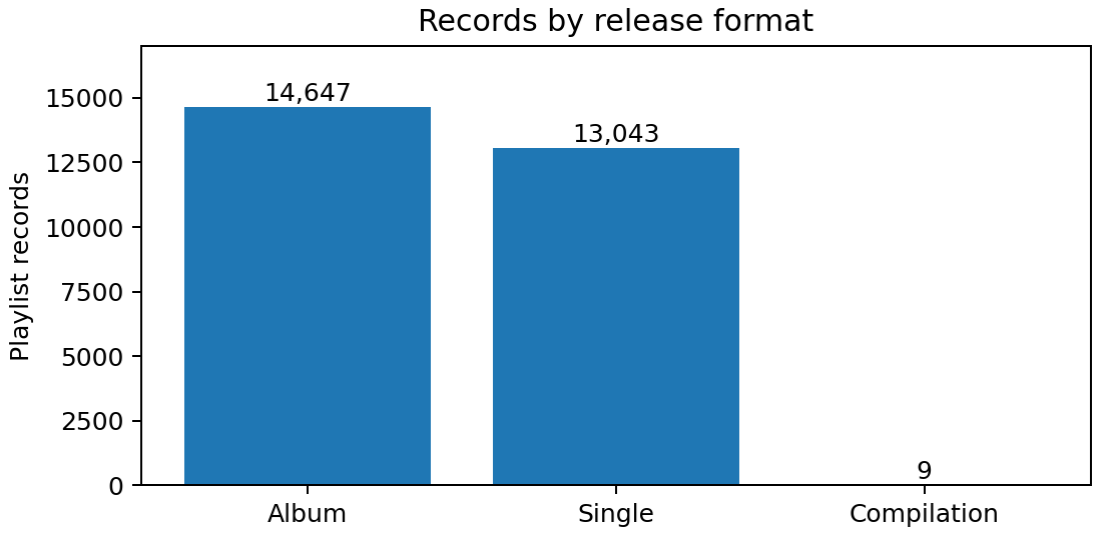


Figure 3. Playlist-record counts by release-format metadata.

6.1 What the format split suggests

The mix is relatively balanced between album and single categories, with album-associated records exceeding single-associated records by 1,604 observations. Compilations are negligible in this table. This is useful for catalogue and metadata reporting: both album and single contexts are materially represented, so a workflow focused exclusively on one release type would overlook a substantial portion of the observed records.

6.2 Album size versus popularity

The dashboard includes a scatter plot of total tracks in a release against average Spotify popularity and a derived Album Size Impact Index of 3.26. However, the report export does not provide the index formula or sufficient numeric detail to independently validate the direction and magnitude of the relationship. The value 3.26 should therefore be treated as a dashboard-specific metric until its formula, units, and interpretation are documented. It must not be described as a causal effect of album size on popularity.

6.3 Top album sizes by record count

The album-size chart is a distribution of playlist-record observations grouped by release track count. It answers how many records are associated with releases of a given size; it does not by itself establish which album size listeners prefer, which releases are more successful, or whether a larger album produces better performance. If the objective is album-structure analysis, the visual should be filtered to album-type records and its title should make clear that the measure is a record count.

6.4 Release-planning implication

Use release format as a descriptive segmentation field, not as a stand-alone decision rule. To compare formats fairly, the next iteration should aggregate at the unique-track or unique-release level, control for repeated appearances, and report popularity distributions (median and interquartile range as well as mean). A release-level comparison should also account for the number of tracks and release age.

7. Duration, Popularity and Time-Based Patterns

The dashboard reports an average song duration of 3.09 minutes. It also contains a duration distribution, duration-versus-popularity scatter plot, duration-versus-playlist-position scatter plot, and average popularity over time. These visuals are useful exploratory tools, but the PDF export does not expose enough exact point-level values to support claims about a precise correlation coefficient or a statistically significant trend.

7.1 Duration profile

An average duration of 3.09 minutes provides a concise description of track length in the observed records. The dashboard groups tracks into short (under 3 minutes), medium (3–5 minutes), and long (over 5 minutes) categories. Because exact category totals are not printed legibly in the exported report, this paper does not state precise percentages for these bins. The bins are useful for exploration, but the cut-offs are analytical choices rather than universal definitions of short or long music.

7.2 Duration versus popularity

A scatter plot can help reveal whether very short or very long tracks tend to carry different popularity scores. The visual alone cannot establish that duration drives popularity, and a cloud of points may be affected by repeated observations of the same track. A robust follow-up would compute the relationship at the unique-track level, inspect outliers, and compare popularity distributions across duration bins. Report the sample size and correlation method if a correlation statistic is added.

7.3 Duration versus rank

The position field is ordinal: position 1 is a stronger chart position than position 50. Therefore, the average position is a descriptive summary, not a conventional performance score where higher is better. If the chart plots duration against average position, a lower average position indicates a better typical rank. Using SUM(position) would be misleading because the total depends heavily on how many times a track appears; average position is the more interpretable aggregation for repeated observations.

7.4 Popularity over time

The executive page plots average Spotify popularity across dates. This is useful for monitoring the movement of a platform-provided score through the period, but a line chart should not be interpreted as a direct trend in streams or audience sentiment. The missing date of 1 March 2025 and the repeated appearance of tracks can affect the time series. A stronger version would include a clearly defined daily average, a rolling average, date completeness checks, and a note about the excluded date.

7.5 Suggested statistical extensions

Calculate Pearson or Spearman correlation only after choosing the appropriate unit of analysis and checking assumptions; use Spearman when assessing monotonic association with rank. Compare median popularity by duration category and release type. If making a claim about changes over time, quantify the change and uncertainty rather than relying only on visual inspection.

8. Business Implications and Recommendations

The project is most useful as a decision-support report for editorial, content operations, and catalogue-analysis teams. The recommendations below follow from the descriptive results while keeping the distinction between observed patterns and untested explanations explicit.

8.1 Content compliance and editorial review

Because 56.27% of all observations and 62.47% of Top 10 observations are flagged explicit, content classification should be visible in editorial review workflows. A practical checklist is: confirm the exact recording/version; inspect the explicit flag; verify market and audience requirements; review the clean edit if one exists; record the decision and reviewer; and re-check the metadata when catalogue records change. The dashboard should support prioritization, not automatically label content as suitable or unsuitable.

8.2 Playlist pitching

A pitch can include the track's release format, duration, rank history, and popularity score as contextual metadata. However, the current findings do not justify promising that a particular format, duration, or explicit status will increase placement. Pitch materials should focus on the track's fit with the target playlist, campaign objectives, audience, and verified performance data.

8.3 Localization and audience suitability

A France-market playlist context makes local editorial requirements important, but this dataset does not contain listener demographics, regional language comprehension, cultural-response measures, or survey feedback. It therefore cannot establish what French listeners find culturally sensitive. Localization decisions should combine metadata analysis with market-specific editorial guidance and, where needed, qualitative review or audience research.

8.4 Release-format strategy

The near-balanced album/single record mix suggests both release contexts merit consideration in reporting. It does not prove that albums outperform singles. Before advising a release strategy, compare unique tracks and releases, control for track repetition, release age, artist popularity, promotion, and the number of tracks per release. Use the Album Size Impact Index only after its formula is documented and validated.

8.5 Dashboard improvements

Priority	Improvement	Why it matters
High	Show denominators and exact labels on every percentage chart.	Makes values auditable and prevents rounded labels from being mistaken for exact counts.
High	Add a unique-track view beside the record-weighted view.	Reduces the effect of tracks appearing repeatedly across dates.
High	Document the Album Size Impact Index formula.	Makes the custom metric reproducible and interpretable.
High	Recalculate exact explicit shares for Top 10, Top 25, and Top 50.	Avoids unsupported interpretation of visually unclear rank-tier bars.
Medium	Add a data-quality note for 2025-03-01 exclusion.	Makes the time-series gap transparent.
Medium	Clarify popularity and rank aggregation in chart titles.	Prevents confusion between average rank, total rank, and popularity.

9. Limitations, Ethics and Future Work

9.1 Limitations

First, the dataset is observational and descriptive. It cannot identify causal effects of explicit content, duration, or release format. Second, repeated playlist records mean that record-weighted shares are not equivalent to unique-song shares. Third, Spotify popularity is a platform-generated score, not a direct stream count or revenue measure [1]. Fourth, the explicit flag is metadata and may not substitute for a full content review. Fifth, the 1 March 2025 date was excluded because its records contained conflicting positions; the 2 February 2025 record with a missing song and zero duration also required cleaning. Sixth, exact numerical values for some visuals—particularly rank-tier shares, duration-bin counts, and the album-size index—are not legible or fully defined in the exported report. This paper intentionally does not fabricate those figures.

9.2 Ethical and responsible interpretation

Explicit-content classification should be used for appropriate editorial context and compliance, not as a proxy for the quality, artistic merit, or suitability of an artist as a person. A flag-based report should be applied consistently, with human review for high-impact decisions. Any audience-facing or brand-safety decision should consider the specific campaign and market requirements rather than assuming that all explicit-flagged material is unsuitable.

9.3 Future work

The next iteration should: (1) obtain stable track and release identifiers; (2) calculate unique-track and unique-release metrics; (3) validate exact rank-tier percentages from the cleaned source; (4) document custom DAX measures, especially the Album Size Impact Index; (5) calculate median and distributional statistics for popularity and duration; (6) quantify time trends with rolling averages and explicit date-completeness checks; (7) incorporate streams or other performance measures only where properly licensed and available; and (8) add a documented manual-review protocol for content classification where the business requirement calls for it.

9.4 Conclusion

This analysis provides a structured descriptive view of France Top 50 playlist records from 18 May 2024 to 27 November 2025. In the cleaned table, explicit-flagged observations represent 56.27% of records, while the Top 10 is 62.47% explicit-flagged and 37.53% clean/not flagged explicit. Album-associated records account for 52.88%, singles 47.09%, and compilations 0.03%; average duration is 3.09 minutes. These findings support better visibility of content flags, release metadata, and playlist-position context. They do not prove audience preference or causal performance effects. The most valuable next step is to improve reproducibility—especially by separating repeated playlist records from unique tracks and by documenting custom measures—so the dashboard can become a more reliable foundation for editorial and operational decisions.

References and Appendix

[1] Spotify for Developers. *Get Track — Web API Reference.* Documentation for track metadata including album type, total track count, duration, explicit flag, and popularity field. Available at: https://developer.spotify.com/documentation/web-api/reference/get-track (accessed 21 October 2026).

[2] Microsoft Learn. *Overview of Power BI reports.* Documentation on interactive, multi-page reports, visualizations, filters, slicers, cross-filtering, and sharing. Available at: https://learn.microsoft.com/en-us/power-bi/create-reports/power-bi-reports-overview (accessed 21 October 2026).

Appendix A. Key reported metrics

Metric	Reported value	Interpretation / note
Final row count	27,699	Playlist-record observations, not unique songs.
Distinct dates	554	18 May 2024–27 November 2025.
Explicit-flag share	56.27%	Across all cleaned records.
Clean/not-flagged share	43.73%	Across all cleaned records.
Clean records per explicit record	0.78	Ratio reported in the dashboard.
Top 10 explicit-flag share	62.47%	Within observations at positions 1–10.
Top 10 clean/not-flagged share	37.53%	Within observations at positions 1–10.
Album records	14,647 (52.88%)	Release-format metadata.
Single records	13,043 (47.09%)	Release-format metadata.
Compilation records	9 (0.03%)	Very small category.
Single-to-album record ratio	0.89	Single record count divided by album record count.
Average duration	3.09 minutes	Across cleaned playlist records.
Album Size Impact Index	3.26	Formula not visible in the exported report; do not interpret as causal.

Appendix B. Terminology

Playlist-record observation: one row representing a track's presence at a position on a date. **Explicit-flagged:** a row whose metadata field is true. **Clean/not flagged explicit:** a row whose metadata field is false; this is not equivalent to a manual lyrical audit. **Popularity score:** Spotify's platform-provided 0–100 field. **Average position:** mean rank across repeated observations; lower values indicate better typical placement. **Record-weighted:** each row contributes once, so tracks appearing on multiple dates contribute repeatedly.

Data transparency statement: This paper is based on the cleaned dataset summary and the five-page Power BI dashboard export available for the project. Values not visible or not defined in those materials are explicitly marked as unavailable rather than estimated.